The disjoint sum of two ordinals under the reverse lexicographic order is a well-order

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Editorial note. The source discharges connectedness with a single word ('Obviously'); the counted proof's substance is the well-foundedness construction, which is given in full.

Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel ordsumlessiswo of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- disjoint sum of two ordinals as a two-piece concatenation
- the reverse lexicographic order on pairs
- well-foundedness witnessed by a least element construction

Lemma 0.1. *$\langle \alpha \oplus \beta, \prec \rangle$ is a well-order, for any ordinals α and β.*

Proof. Obviously $\prec$ is connected on $\alpha \oplus \beta$. To show it is well-founded, fix a non-empty $X \subseteq \alpha \oplus \beta$. Let Y be the subset of X whose second coordinate is as small as possible, i.e. $Y = \{ \langle \gamma, i \rangle \in X \mid (\forall \langle \delta, j \rangle \in X) i \leq j \}$. Now choose the element of Y with smallest first coordinate. □

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